

An Honest Miss: Chinchilla as an Adoption Ramp in a Discontinuity Scan of Tokens per Parameter

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July 2026

analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

A LoRD3 continuous-treatment discontinuity scan of $\log_{10}$ training tokens per parameter over calendar time, run blind on 549 language models (2019–2024) from Epoch AI’s notable-models panel, produces a single significant top discovery: a break centered on 2022-09-22 (LLR 8.687, scan $p \leq 0.01$, robust to background degree and to neighborhood size $k \geq 50$). The known truth — the Chinchilla compute-optimal scaling result, posted 2022-03-29, after which the global median tokens-per-parameter ratio rose from 1.7 to 29.2 — is *not* localized: the best candidate near the announcement ranks 158 of 547 (LLR 1.39). The discovered break is six months late and points the wrong way: a downward local contrast of 1.63 $\log_{10}$ units ($\times 43$) at the Sep/Oct 2022 boundary, where the downstream edge of the Chinchilla adoption wave collides with a measurement artifact — fine-tuned releases whose recorded token counts cover only the fine-tuning run over full base parameters. Nulls are reported as nulls: the placebo test is vacuous (no non-forcing covariates), the density test fails ($p = 0.0041$) but is uninformative for a calendar-time forcing, post-selection 2SLS is weak-instrument, and an independent kink audit rejects a slope-change reading. The verdict is an artifact — an honest miss, correctly diagnosed — with two transportable lessons: diffusion-style treatments need a ramp/slope statistic, not only a level-break statistic, and fine-tune rows must be filtered from tokens-per-parameter panels before any scan.

1 Introduction

On 29 March 2022, Hoffmann et al. posted the “Chinchilla” result: for compute-optimal training, model size and training tokens should scale in equal proportion, implying that the then-standard practice — descended from the earlier scaling-law fits of [7] — left frontier models substantially undertrained on data [6]. Release practice responded within months, and by 2023 the industry had moved past compute-optimality into deliberate over-training of small models for inference economy, exemplified by LLaMA’s trillion-token training runs [12] and rationalized by inference-adjusted scaling laws [10]. Chinchilla itself has since been re-examined: [2] replicate its parametric fit and find inconsistencies in the reported estimates, while the broader quantitative history of training-run inputs is documented by [11] and maintained in Epoch AI’s notable-models dataset [4].

This gives a rare, clean testbed for *automated natural-experiment discovery*: a panel where the event, its date, its sign, and its approximate magnitude are known ex ante. This paper runs the

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natex toolkit’s LoRD3 scan [9, 5] on tokens per parameter over calendar time, blind to the event date, and asks: does the scanner find Chinchilla? The answer is no — and the anatomy of the miss is the contribution. Event-dated diffusion processes enter publication data as ramps, and a local level-break statistic rewards sharp edges, not ramps; dating the cause is an interrupted-time-series problem [1], not a discontinuity-scan problem.

2 Data

The panel is a frozen extract of Epoch AI’s notable-models database [4]: 549 language models released 2019-01 through 2024-12 with non-missing parameter and training-token counts. Columns: release date (the forcing variable, as days since 2019-01-01), $\log_{10}$ tokens per parameter (the treatment), $\log_{10}$ parameters (the outcome), $\log_{10}$ training compute (21% missing) and an open-weights flag (3.6% missing), both excluded from the scan to avoid listwise deletion. The known truth: Chinchilla was posted 2022-03-29 (day ≈ 1183); across that date the global median tokens per parameter moves from 1.69 to 29.17 ($n = 185$ pre, 364 post), a median-based shift of $+1.24 \log_{10}$ units ($+1.12$ on group means, $2.2 \rightarrow 29$; Mann–Whitney $p = 1.6 \times 10^{-7}$ on an earlier extract, medians $2.2 \rightarrow 29.1$, $n = 133/307$). At monthly resolution the shift is a ramp, not a step: window medians run 1.7 (pre-April 2022) $\rightarrow$ 5.7 (mid-April to mid-May) $\rightarrow$ ≈ 28 (mid-May to mid-July 2022).

3 Design and Methods

LoRD3 continuous-treatment scan. Following [5] as implemented in **natex** v0.2.0 [9], the scan fits a background model of the treatment on the forcing variable (OLS, degree 1; degree 2 as robustness), then scores every k -nearest-neighbor neighborhood ($k = 50$) of every data point for the best split of its members into two half-spaces, using a normal likelihood ratio on background residuals with locally estimated variances. The top discovery is the neighborhood-split with the maximum LLR; its p -value comes from a fitted-null Monte Carlo with $q = 99$ replicas and a $+1$ -rank rule, so the minimum attainable p is 0.010 — reported throughout as $p \leq 0.01$, never as an exact value. The specification is treatment = $\log_{10}$ tokens per parameter, forcing = days since 2019-01-01 (forcing influence 1.0), outcome = $\log_{10}$ parameters, no covariates; pipeline **natex study** $\rightarrow$ **natex discover --k 50 --q 99 --seed 0**, with localization detail and robustness via the Python API, seed 0 throughout.

Validation battery, stated as run. The battery’s nulls and refusals are reported, not patched over. (i) Randomization: $p = 0.010$ at both background degrees — the $q = 99$ floor. (ii) Placebo covariates: **placebo_passed** = True but *vacuous* — **placebo_holm** = {}, because the design has no non-forcing covariates to test. (iii) Density: $p = 0.0041$, a *fail*; this is expected and uninformative here, because the forcing is calendar time with strongly nonstationary publication intensity, and McCrary-style manipulation logic [8] does not apply to a time forcing — the test is retained as a falsification check only. (iv) Post-selection effects on $\log_{10}$ parameters (2SLS and Wald at the discovered cutoff) are *advisory*: estimated on the discovery sample with no honest split and instrumented by a partly artifactual break, they must not be read as causal scaling-law parameters. (v) An independent kink-design audit of the same panel (a difference-in-slopes design [3]) cross-checks any slope-change reading.

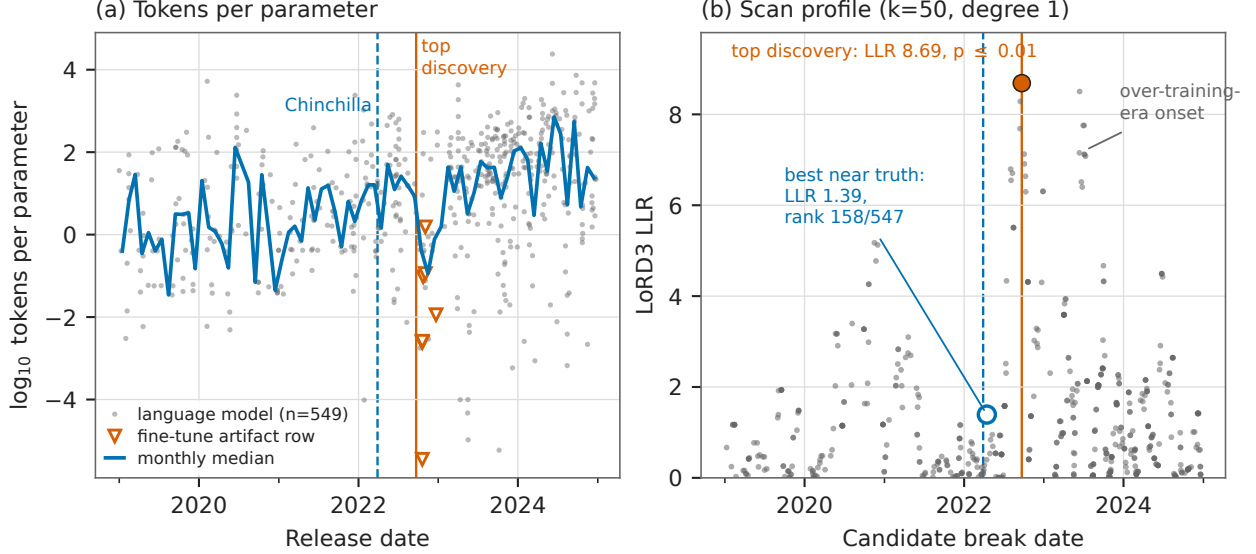


Figure 1: (a) $\log_{10}$ tokens per parameter for all 549 models, with the monthly median (line), the Chinchilla posting date (dashed), the scan’s top discovery (solid), and the seven verified fine-tune token-counting artifact rows (open triangles; Table 1, panel C). (b) LoRD3 scan profile: each candidate neighborhood’s LLR at its center date ($k = 50$, degree 1, $n = 547$ candidates). The top discovery (filled marker, 2022-09-22, LLR 8.69, $p \leq 0.01$) sits at the Sep/Oct 2022 artifact edge; the best candidate within ± 45 days of the truth (open marker, 2022-04-15) scores LLR 1.39, rank 158/547. The 2023-06/07 cluster is the onset of the deliberately over-trained era.

4 Results

The discovery. The scan’s top discovery (Table 1, panel A; Figure 1b) is centered on day 1360 = 2022-09-22 (split normal -1 along the forcing axis): LLR = 8.687 (8.687416 unrounded in the run report), scan $p = 0.010$ — the $q = 99$ Monte Carlo floor, i.e. $p \leq 0.01$ — over 547 candidate neighborhoods from $n = 549$ rows. A degree-2 background reproduces the same top center with LLR 8.836 (randomization $p = 0.01$; null max LLRs 1.8–7.6 against observed 8.69/8.84), and $k = 60$ moves it only to 2022-08-15; but $k = 30/40$ latch onto a 2020-10 composition boundary instead (Table 1, panel B), so localization is stable only for $k \geq 50$.

The known truth is not localized. Within ± 45 days of 2022-03-29 the best candidate neighborhood is centered 2022-04-15 with LLR 1.39 — rank 158 of 547. The reason is visible in Figure 1a: at daily resolution the Chinchilla shift enters the published record as a 2–3 month adoption ramp, and a local level-break statistic rewards sharp edges, not ramps.

What the discovered break actually is. The top neighborhood splits exactly at the Sep/Oct 2022 boundary into group A (2022-06-21 to 2022-09-22, $n = 25$, median tokens/param 13.6 — the post-Chinchilla adoption wave) versus group B (2022-10-03 to 2022-12-22, $n = 25$, median 0.4): a *downward* local group-mean contrast of 1.63 $\log_{10}$ units ($\times 43$), against the known *upward* global shift of +1.12 $\log_{10}$ (+1.24 median-based). Local-linear first-stage fits jump -1.7 to -2.0 $\log_{10}$ at bandwidths of 120–240 days. Group B is dominated by fine-tuned and continued-pretraining releases whose Epoch token field records only fine-tuning tokens over full base parameters (Table 1,

Table 1: Chinchilla scan: numbers of record (**natex** v0.2.0, seed 0).

Quantity	Value	Quantity	Value
<i>Panel A: top discovery ($k = 50$, $q = 99$, degree 1; $n = 549$, 547 candidates)</i>			
center	2022-09-22 (day 1360)	LLR	8.687
scan p	0.010 = floor; $p \leq 0.01$	local contrast	$-1.63 \log_{10} (\times 43, \text{down})$
degree-2 center	2022-09-22, LLR 8.836	first stage	$-1.7 \text{ to } -2.0 \log_{10} (\text{bw } 120\text{--}240\text{d})$
<i>Panel B: k-sensitivity and truth localization</i>			
$k = 30$	2020-09-29 (LLR 7.2)	$k = 40$	2020-10-21 (LLR 9.0)
$k = 50$	2022-09-22 (LLR 8.7)	$k = 60$	2022-08-15 (LLR 9.8)
truth $\pm 45\text{d}$ best	2022-04-15, LLR 1.39	truth rank	158 / 547
global shift (true)	$+1.12 \log_{10} (2.2 \rightarrow 29, \text{up})$	medians	$1.69 \rightarrow 29.17$ ($n = 185/364$)
<i>Panel C: verified fine-tune artifact rows (tokens/param; group B median 0.4 vs. A 13.6)</i>			
Flan-PaLM 540B	0.0026	U-PaLM	0.0024
LMSI-PaLM	3.6×10^{-6}	OPT-IML 175B	0.011
Tk-Instruct	0.095	BLOOMZ-176B	0.114
mT0-13B	1.54		
<i>Panel D: diagnostics and post-selection effects (advisory)</i>			
randomization	$p = 0.010$ (deg. 1 and 2)	placebo	passed, <i>vacuous</i> (no covariates)
density	$p = 0.0041$, fails [†]	kink audit	rejected; $t \in [-0.91, +1.48]$
2SLS $\hat{\tau}$	1.017 (se 0.900), weak	CI	$[-0.747, 2.781]$, 1st-st. $t = 1.89$
Wald $\hat{\tau}$	0.557 (se 0.257)	CI	$[0.053, 1.062]$, 1st-st. $t = 4.01$

Notes: [†]expected and uninformative for a calendar-time forcing with nonstationary publication intensity (McCrary manipulation logic does not apply). Scan p -values from a $q = 99$ fitted-null Monte Carlo with a +1-rank rule; 0.010 is the minimum attainable. Effects are estimated on the discovery sample with no honest split. Artifact rows verified in the source panel: the token field records fine-tuning tokens only, over full base parameters.

panel C) plus small academic models. The top discovery is therefore the adoption wave’s downstream edge colliding with a token-accounting artifact, ≈ 6 months after the event and with the opposite sign. The secondary discovered cluster (2023-06-13 to 2023-07-06, LLR 6.7–8.5) is the onset of the deliberately over-trained era [12, 10]; the December-2022 monthly median is already 148 tokens/param, and the median of the top-25 centers is day 1371 = 2022-10-03.

Effects and cross-checks, all null or advisory. Post-selection effects on $\log_{10}$ parameters at the discovered cutoff (Table 1, panel D): the 2SLS estimate is weak-instrument (first-stage $t = 1.89$; CI $[-0.747, 2.781]$); the Wald estimate is 0.557 (CI $[0.053, 1.062]$, first-stage $t = 4.01$). Both are discovery-sample estimates with no honest split, instrumented by a partly artifactual break — diagnostics, not causal scaling-law parameters. The independent kink audit *rejects* a Chinchilla regime change read as a slope change: a sign-unstable null with t from -0.91 to $+1.48$ across bandwidths.

5 Caveats and Conclusion

Five caveats bound the claims. First, the value of this exercise is purely methodological and diagnostic: no causal adoption claim is licensed; the scan is a regime-edge detector, and dating the cause of a diffusion process needs an interrupted-time-series or difference-in-differences framing [1] plus publication-lag domain knowledge. Second, the scan $p = 0.010$ is the minimum attainable with $q = 99$ replicas and must be reported as $p \leq 0.01$, not as an exact value. Third, the density test fails ($p = 0.0041$) but is uninformative for a calendar-time forcing with nonstationary publication intensity, and the placebo pass is *vacuous* (`placebo_holm = {}`; no non-forcing covariates to test). Fourth, the post-selection 2SLS/Wald effects are estimated on the discovery sample with no honest split and are instrumented by a partly artifactual break; they must not be read as causal scaling-

law parameters. Fifth, “robust to k ” holds only for $k \geq 50$: at $k = 30/40$ the scan latches onto a 2020-10 composition boundary (tiny academic LMs against BERT-era corpus models), so the robustness claim must be stated conditionally.

The verdict is an artifact — an honest miss, correctly diagnosed: the scan found real, significant structure whose two regimes bracket the truth, but it dated the adoption wave’s sharp downstream *edge*, not the event, which entered the record as a ramp. Two lessons transport beyond this dataset. For method builders: scan batteries aimed at event-dated diffusion treatments need a ramp/slope statistic — a kink-style scan in the sense of [3] — alongside the level-break statistic, or announcements will systematically lose to their own adoption edges. For users of the Epoch panel [4]: fine-tuned and continued-pretraining rows manufacture a spurious low-tokens-per-parameter cluster in late 2022 and must be filtered before tokens-per-parameter data are used for scaling-law or event-study claims [6, 2].

Reproducibility. All estimates: `natex` v0.2.0 [9], seed 0, on a frozen extract of the Epoch panel [4] (not committed). Figure 1 regenerates deterministically from `figures/make_fig.py` (committed), which asserts the headline numbers of record before drawing.

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